

Instrumental Variables

Applied Econometrics in Practice

ECO372H1F

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What this lecture is about

The biggest problem in applied microeconomics is that we usually cannot run an experiment.

We want to know: does D *cause* Y ?

But in the real world, the variables that determine D are tangled up with the variables that determine Y . Running OLS gives a number, but that number is not the causal effect.

Instrumental variables (IV) is a method for getting causal effects out of observational data, by finding a clever third variable.

This lecture is the most important technique in this course.

- Block 1.** The endogeneity problem.
- Block 2.** The IV idea and the Wald estimator.
- Block 3.** Linear IV: reduced form, first stage, 2SLS.
- Block 4.** Threats to validity: independence and exclusion.
- Block 5.** Heterogeneous effects and LATE.
- Block 6.** The four response types: compliers and friends.
- Block 7.** Diagnostics: weak instruments, overidentification.
- Block 8.** Applications and worked examples.

What is endogeneity, and why does it ruin OLS?

The setup

We have a sample of individuals. For each person i , we observe:

- An outcome Y_i (e.g., wage, test score, health).
- A treatment D_i (e.g., college attendance, medication, programme participation).
- Maybe some covariates W_i .

The causal model we believe in:

$$Y_i = \alpha + \beta D_i + \varepsilon_i.$$

We want to estimate β — the causal effect of D on Y .

The simplest tool is OLS: run a regression of Y on D . The slope coefficient is our estimate of β .

What's in the error term?

The error term ε_i is a black box that contains *everything that affects Y_i except D_i* .

Examples of things in ε_i :

- Innate ability and motivation.
- Family background.
- Health.
- Luck.
- Things we don't know enough to even name.

The error term is unobserved, by definition.

When OLS works

OLS gives a consistent estimate of β when:

$$\text{Cov}(D_i, \varepsilon_i) = 0.$$

In words: the treatment D_i is uncorrelated with everything else that affects Y_i .

This is sometimes called *exogeneity* of D_i .

If this condition holds, then variation in D across people is unrelated to anything else that matters for Y , and the regression slope picks up the pure effect of D .

When OLS fails

If $\text{Cov}(D_i, \varepsilon_i) \neq 0$, OLS does not recover β . We say D_i is **endogenous**.

The bias formula:

$$\text{plim } \hat{\beta}_{\text{OLS}} = \beta + \frac{\text{Cov}(D_i, \varepsilon_i)}{\text{Var}(D_i)}.$$

Three things to notice:

- 1 The bias does not shrink as the sample grows. More data does not save you.
- 2 The *sign* of the bias matches the sign of $\text{Cov}(D, \varepsilon)$.
- 3 The *magnitude* of the bias depends on how strongly D tracks the unobservables.

Three sources of endogeneity

Endogeneity is the disease. To diagnose it, look for one of three causes:

- ① **Omitted variable bias:** an unobserved factor causes both D and Y .
- ② **Measurement error in D :** we don't observe the true D , only a noisy version.
- ③ **Simultaneous causality:** D and Y cause each other.

We'll spend the next several slides on each.

Source 1: Omitted variable bias — the idea

A variable affects both D and Y , but we don't include it in our regression. It hides in the error term. Because the omitted variable affects D , and the omitted variable is part of ε , we get $\text{Cov}(D, \varepsilon) \neq 0$. This is the most common source of endogeneity in applied work.

Omitted variable bias — example: schooling and wages

We want to know: does an extra year of school raise wages?

We run:

$$\log W_i = \alpha + \beta \cdot \text{Education}_i + \varepsilon_i.$$

What's hiding in ε_i ?

- Innate ability.
- Motivation and grit.
- Family connections.

These are all unobserved (we don't have an “ability” column in our dataset).

Schooling and wages: why OLS is biased

Innate ability affects both education and wages:

- More able people stay in school longer.
- More able people earn more, even at the same education level.

So ability is positively correlated with education *and* positively correlated with wages.

Result: $\text{Cov}(\text{Education}, \varepsilon) > 0$, so OLS *overestimates* the return to schooling.

The estimated coefficient mixes the true causal effect of education with the wage premium of ability.

Schooling and wages: how big is the problem?

The classical OLS estimate of the return to schooling is roughly 10% per year of education.

But: how much of that is the actual effect of learning, and how much is just that smarter people both stay in school and earn more?

This is the central question of an entire literature in labour economics. We'll return to it many times in this lecture.

Omitted variable bias — example: smoking and health

We want to know: does smoking cause lung cancer?

Compare smokers and non-smokers. Smokers have higher rates of lung cancer.

But: smokers also differ from non-smokers in many other ways.

- Diet quality.
- Exercise habits.
- Stress levels.
- Whether they visit doctors regularly.

All of these affect both the decision to smoke *and* the probability of cancer.

Smoking: which way does the bias go?

“Health consciousness” is the hidden variable.

Health-conscious people are less likely to smoke *and* less likely to develop cancer (for reasons unrelated to smoking).

So smoking and the error term are positively correlated. OLS overestimates the cancer risk from smoking.

(Smoking really does cause cancer, but the unadjusted correlation overstates it. RCTs would be ideal but are ethically impossible.)

Omitted variable bias — example: police and crime

We want to know: does hiring more police reduce crime?

Run a regression of crime rates on police-per-capita across cities.

What do we find? Often, more police is correlated with *more* crime.

Does that mean police cause crime?

Police and crime: the omitted variable

No. The omitted variable is *expected crime*.

Cities anticipate high crime → hire more police.

Cities with high expected crime have actual high crime.

So expected crime is correlated with both the regressor (police) and the outcome (crime). OLS captures the wrong direction.

This is the classic “reverse causality” or “selection on outcomes” story — a special case of omitted variables.

Omitted variable bias — example: class size

We want to know: do smaller classes improve test scores?

Regress test scores on class size. What do we typically find?

The coefficient is often *positive*: smaller classes look like they *hurt* test scores.

Class size: why OLS is misleading

Schools assign small classes to:

- Struggling students who need extra attention.
- Special-needs students.

These students have lower test scores for reasons unrelated to class size.

The hidden variable is “student needs.” It pushes class size down and test scores down.

$\text{Cov}(\text{class size}, \varepsilon) > 0$, so OLS gives an upwardly biased estimate. The OLS coefficient says smaller classes “cause” lower scores, when really the small classes were assigned to weaker students to begin with.

Omitted variable bias — the general principle

Omitted variable bias happens when:

- 1 Some variable U is not in your regression.
- 2 U affects D .
- 3 U affects Y (other than through D).

The OLS coefficient on D then mixes the true effect of D with the effect of U that runs through the D -channel.

You can sometimes fix this by including U as a control — but only if you can measure U . Often you cannot. That is the problem.

Source 2: Measurement error in D — the idea

Suppose the true treatment is D_i^* , but we only observe a noisy version:

$$D_i = D_i^* + u_i,$$

where u_i is random measurement noise.

Examples:

- Self-reported years of schooling. People misremember.
- Self-reported income. People round, lie, or forget.
- Self-reported smoking status. Some smokers say they don't.

This noise creates endogeneity even if the underlying treatment is exogenous.

Measurement error — the algebra

The structural model uses the true variable:

$$Y_i = \alpha + \beta D_i^* + \varepsilon_i.$$

But we run OLS using the noisy $D_i = D_i^* + u_i$. Substituting:

$$Y_i = \alpha + \beta D_i + (\varepsilon_i - \beta u_i).$$

Look at the new error: it contains $-\beta u_i$.

But u_i is part of D_i . So D_i is correlated with the new error term by construction.

$\text{Cov}(D_i, \varepsilon_i - \beta u_i) \neq 0$. We have endogeneity.

Measurement error — the bias

The plim of OLS is:

$$\text{plim } \hat{\beta}_{\text{OLS}} = \beta \cdot \frac{\text{Var}(D^*)}{\text{Var}(D^*) + \text{Var}(u)}.$$

The factor is strictly less than 1. So $\hat{\beta}_{\text{OLS}}$ is biased *toward zero*.

This is called **attenuation bias**.

Crucially: adding random noise to your regressor does NOT just inflate standard errors. It *flattens the slope*. The estimated effect looks smaller than the truth.

Measurement error — example: returns to schooling

Suppose the true return to schooling is 10%.

Suppose education is measured with noise such that 25% of the variance in observed education is noise.

Then the OLS coefficient will be:

$$0.10 \times \frac{0.75}{1} = 0.075.$$

We would estimate a 7.5% return when the truth is 10%. The estimate is too low.

This is one reason some IV estimates of the return to schooling come out *higher* than OLS — they correct for attenuation.

Measurement error — example: smoking

Suppose we ask people: how many cigarettes did you smoke yesterday?

People underreport. Some refuse to admit they smoked at all.

If true smoking is the D^* and reported smoking is D , then D is a noisy version of D^* .

OLS of, say, lung capacity on reported smoking will give an attenuated estimate of the true effect.

This is a common issue in health economics whenever the treatment is self-reported.

Source 3: Simultaneous causality — the idea

Sometimes D causes Y , but Y also causes D .

The two variables are determined jointly. A regression cannot disentangle which way the arrow runs.

This is sometimes called *reverse causality* or *joint determination*.

It is endogeneity because the error in the Y -equation feeds back into D .

Simultaneity — example: supply and demand

Suppose we regress quantity sold Q on price P for some good.

What does the coefficient mean?

- A demand curve says: when price rises, quantity demanded falls. Negative slope.
- A supply curve says: when price rises, quantity supplied rises. Positive slope.

The data we observe is a sequence of equilibrium points where supply equals demand. The slope we get is neither the supply nor the demand curve.

We need an instrument to separate them.

Simultaneity — the algebra

Suppose:

$$Q_i = \alpha_D - \beta_D P_i + \varepsilon_i^D \quad (\text{demand})$$

$$Q_i = \alpha_S + \beta_S P_i + \varepsilon_i^S \quad (\text{supply})$$

A demand shock ε_i^D shifts demand $\rightarrow$ both P and Q rise.

So in the demand equation, $\text{Cov}(P, \varepsilon^D) > 0$. Regressing Q on P doesn't give the demand slope; it gives some mixture.

Simultaneity — example: police and crime, revisited

Earlier we discussed police and crime as an OVB problem (expected crime).

It's also a simultaneity problem:

- More police $\rightarrow$ less crime (the effect we want).
- More crime $\rightarrow$ city hires more police (reverse channel).

Even if there is no omitted variable, the simultaneous determination means OLS cannot recover the first effect cleanly.

These categories overlap. Don't worry too much about labelling — worry about identifying the channel and fixing it.

Simultaneity — example: democracy and growth

Does democracy cause economic growth?

Compare countries: democracies are richer.

But:

- Democracy might cause growth (good institutions $\rightarrow$ investment $\rightarrow$ growth).
- Growth might cause democracy (richer societies demand political rights).

Cross-country regressions don't separate these. This is one of the canonical applications of IV in macroeconomics (Acemoglu, Johnson, Robinson 2001 used colonial settler mortality as an instrument).

Three sources of endogeneity: summary

- ① **Omitted variable bias.** An unobserved U affects both D and Y .
- ② **Measurement error in D .** We observe a noisy version of the true treatment, leading to attenuation.
- ③ **Simultaneous causality.** Y and D are jointly determined.

In each case, $\text{Cov}(D, \varepsilon) \neq 0$ and OLS gives a biased estimate.

The bias does not vanish with more data. Sample size doesn't help.

We need a different strategy. That's what IV is for.

What is an instrument, and why does it solve the problem?

The big idea

The problem with OLS: there's too much variation in D , and some of it is confounded.

The IV solution: don't use all the variation in D . Use only the part of the variation that we can prove is exogenous.

How? Find a third variable Z that:

- Pushes D around (so D responds to Z),
- but is unrelated to anything else that affects Y .

The slice of variation in D that comes from Z is clean. Use only that slice.

This is what an **instrument** does.

An analogy: separating dye in water

Imagine D is water with red and blue dye mixed in. The red dye is the causal variation; the blue dye is the confounding.

You can't see which drops are red and which are blue.

But you have a magnet that pulls only the red drops. Apply the magnet, and the water moves — only because of the red drops.

That motion is what IV uses. The magnet is the instrument.

The two properties of an instrument

For Z to serve as an instrument for D in a regression of Y on D , it must satisfy:

1. Relevance. Z actually moves D .

$$\text{Cov}(Z, D) \neq 0.$$

2. Exogeneity. Z does not affect Y through any channel other than D , and is not correlated with confounders of Y .

$$\text{Cov}(Z, \varepsilon) = 0.$$

We will spend a lot of time on these two conditions. They are the heart of the technique.

The slogan

Slogan

Exogenous variation in Z changes the endogenous D . Any resulting change in Y reflects only the causal effect of D .

Two halves:

- “Variation in Z changes D ” = relevance.
- “Any resulting change reflects only D ” = exogeneity.

Keep this slogan in mind. Every IV argument has this shape.

Running example: the oversubscribed course

A motivating example we'll use throughout the lecture.

Setting. A popular course can take 100 students but 300 sign up. The registrar uses a lottery to assign seats.

- $D_i \in \{0, 1\}$: takes the course.
- Y_i : starting salary years later.
- $Z_i \in \{0, 1\}$: won the lottery.

We want to know: does the course raise wages?

Why OLS fails for the course example

If we regress wages on whether you took the course, we get a biased estimate.

Why? Students who took the course differ from students who didn't:

- Maybe they're more ambitious.
- Maybe they have richer parents who connected them to the field.
- Maybe they're already in a job-search pipeline.

All of these are in the error term. All of them are positively correlated with course-taking AND with wages. OLS overstates the effect.

This is omitted variable bias.

Why the lottery is a good instrument

Check the two conditions:

Relevance. Lottery winners are much more likely to take the course (because they have a seat). So $\text{Cov}(Z, D) > 0$. ✓

Exogeneity.

- Lottery assignment is random by design. It's not correlated with ambition, parental wealth, or anything else.
- Winning the lottery doesn't affect wages *except by making it easier to take the course*.

So $\text{Cov}(Z, \varepsilon) = 0$. ✓

The lottery is a valid IV.

Course example: what does IV give us?

The IV estimator will compare:

- Wages of lottery winners vs lottery losers.
- Scaled by the difference in course-taking between winners and losers.

We'll work out the formula in the next block. For now, the intuition:

- The lottery gives us a slice of variation in course-taking that we know is clean.
- That slice is enough to identify the causal effect.

Why we need DAGs

We're about to use a tool from causal inference called **DAGs** — directed acyclic graphs.

A DAG is just a picture of which variables cause which.

We use them because they make the IV assumptions visually obvious, and because they help us spot when assumptions fail.

Three slides on how to read them, then we'll use them to picture the IV setup.

Reading a DAG: nodes and arrows

Nodes. Each circle is a variable. Could be observed or unobserved.

Arrows. Each arrow is a direct causal effect from one variable to another.

No cycles. You can't follow arrows in a loop back to the same variable. (That would be simultaneity, which DAGs handle differently.)

The whole story of a causal model is in the arrows.

Reading a DAG: back-door paths

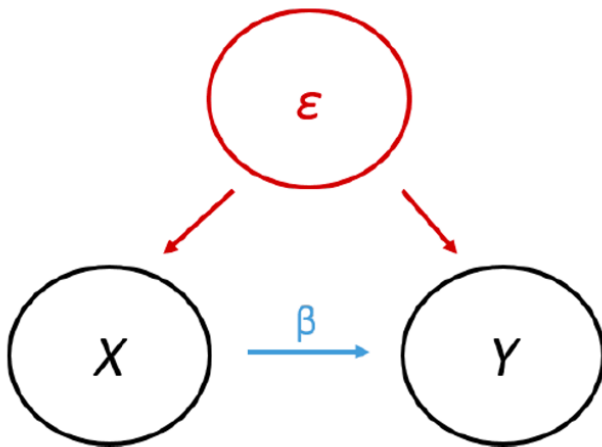
Two variables can be correlated for two reasons:

1. **Direct causation.** An arrow from one to the other.
2. **Common cause.** They share an ancestor. This creates a “back-door” path.

OLS averages over all paths between D and Y . It cannot distinguish a direct effect from a back-door correlation.

That's exactly why OLS fails under endogeneity — there's a back-door path through an unobserved confounder.

The endogeneity DAG



What this DAG is showing

Three nodes:

- X (treatment, our endogenous variable).
- Y (outcome).
- ε (unobserved confounder, in red).

The blue arrow $X \rightarrow Y$ is the causal effect β we want.

The two red arrows $\varepsilon \rightarrow X$ and $\varepsilon \rightarrow Y$ are the problem.

There are now two paths from X to Y :

- Direct: $X \rightarrow Y$ (the causal effect).
- Back-door: $X \leftarrow \varepsilon \rightarrow Y$ (the confounding).

OLS averages both. The OLS coefficient is not β .

Mapping the schooling example onto the DAG

- X = years of education.
- Y = log wage.
- ε = innate ability (unobserved).

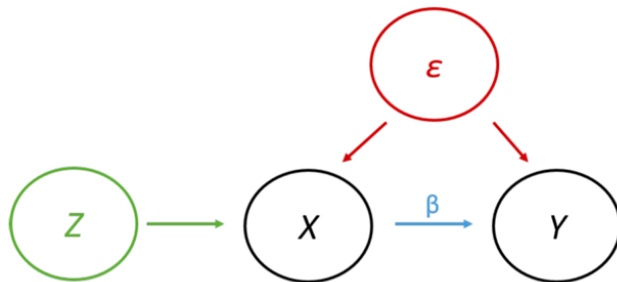
Arrows:

- Ability $\rightarrow$ education (more able students stay in school longer).
- Ability $\rightarrow$ wage (more able workers earn more).
- Education $\rightarrow$ wage (the causal return to schooling).

Regressing wage on education recovers the direct arrow *plus* the back-door through ability.

That's why OLS overstates the return to schooling.

The IV DAG



What's new in the IV DAG

We add a new node Z (in green) and one new arrow: $Z \rightarrow X$.

Three things are critical about Z :

- ① There's an arrow from Z to X . (Relevance.)
- ② There's *no* arrow from ε to Z . (Independence.)
- ③ There's *no* arrow from Z directly to Y . (Exclusion.)

These three features are exactly what make Z a valid instrument.

Why the IV DAG works

With this structure, there is a clean path from Z to Y :

$$Z \rightarrow X \rightarrow Y.$$

And no contamination:

- Z is not a child of ε , so no confounding from above.
- Z doesn't sneak into Y through another route.

The only way variation in Z can move Y is through X . So if we see Y respond to Z , the size of that response (scaled by how much X also responded) tells us the causal effect of X on Y .

Mapping the lottery example onto the IV DAG

- Z = lottery outcome (won or lost).
- X = took the course.
- Y = starting salary.
- ε = ambition, family connections (unobserved).

The arrows:

- Lottery $\rightarrow$ course-taking (relevance).
- Course-taking $\rightarrow$ salary (the effect we want).
- Ambition $\rightarrow$ course-taking (omitted-variable bias channel).
- Ambition $\rightarrow$ salary (omitted-variable bias channel).

No arrow from ambition to the lottery (it's random).

No arrow from the lottery directly to salary (it only matters because of the course).

Block 2 wrap-up

- OLS fails when the regressor is endogenous.
- IV solves the problem by isolating clean variation.
- An instrument Z must satisfy relevance and exogeneity.
- The IV DAG shows the structure: $Z \rightarrow X \rightarrow Y$, with no contamination.

Next: how do we actually compute the IV estimate?

The Wald estimator, reduced form, first stage, 2SLS.

Deriving the IV estimator: setup

Start with the structural equation:

$$Y_i = \alpha + \beta D_i + \varepsilon_i.$$

We want to solve for β in terms of things we can measure.

We have an instrument Z_i that satisfies:

- Relevance: $\text{Cov}(Z, D) \neq 0$.
- Exogeneity: $\text{Cov}(Z, \varepsilon) = 0$.

Watch what happens when we take $\text{Cov}(\cdot, Z_i)$ on both sides of the structural equation.

Deriving the IV estimator: covariance trick

Take covariance with Z_i on both sides:

$$\text{Cov}(Z_i, Y_i) = \text{Cov}(Z_i, \alpha + \beta D_i + \varepsilon_i).$$

Constants have zero covariance with anything, and covariance is linear:

$$\text{Cov}(Z_i, Y_i) = \beta \cdot \text{Cov}(Z_i, D_i) + \text{Cov}(Z_i, \varepsilon_i).$$

By exogeneity, the last term is zero:

$$\text{Cov}(Z_i, Y_i) = \beta \cdot \text{Cov}(Z_i, D_i).$$

By relevance, we can divide both sides by $\text{Cov}(Z, D)$:

$$\beta = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}.$$

The Wald estimand

The formula

$$\beta = \frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, D_i)}$$

is called the **Wald estimand**.

Substituting sample covariances for population covariances gives the IV estimator.

The estimator is the ratio of two simple objects:

- How much Y moves when Z moves (numerator).
- How much D moves when Z moves (denominator).

Wald with a binary instrument

When Z is binary (e.g., the lottery), the Wald formula becomes especially clean:

$$\beta_{IV} = \frac{\mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0]}{\mathbb{E}[D \mid Z = 1] - \mathbb{E}[D \mid Z = 0]}.$$

The numerator: difference in average Y between $Z = 1$ and $Z = 0$.

The denominator: difference in average D between $Z = 1$ and $Z = 0$.

With a binary instrument, computing IV is literally just two means in the numerator and two means in the denominator.

Naming the two pieces

The numerator and denominator have important names.

Reduced form.

$$Y_i = \kappa + \rho Z_i + v_i.$$

The slope $\rho = \text{Cov}(Z, Y)/\text{Var}(Z) = \mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]$ (binary case).

First stage.

$$D_i = \mu + \pi Z_i + \eta_i.$$

The slope $\pi = \text{Cov}(Z, D)/\text{Var}(Z) = \mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0]$ (binary case).

The Wald estimator is just:

$$\beta_{\text{IV}} = \frac{\rho}{\pi}.$$

What the reduced form means

The reduced form is the regression of the *outcome* on the *instrument*.

It tells you: how much does the outcome change when the instrument changes?

In an experiment with imperfect compliance (like the lottery), the reduced form is also called the **intent-to-treat (ITT)** effect:

- Among people offered the treatment ($Z = 1$), what's the average outcome?
- Among people not offered ($Z = 0$), what's the average outcome?
- $ITT =$ the difference.

ITT measures the effect of the *offer*, not the effect of the *treatment*.

What the first stage means

The first stage is the regression of D on Z .

It tells you: how much does D change when Z changes?

In the lottery example:

- Among lottery winners, what fraction took the course?
- Among lottery losers, what fraction took the course?
- First stage = the difference.

The first stage measures the *strength* of the instrument — how much it actually moves the treatment.

If the first stage is small or zero, the instrument is weak. More on this later.

Numerical example: the lottery

Suppose we have:

- Lottery winners: 70% take the course; their average salary is \$60,000.
- Lottery losers: 20% take the course; their average salary is \$55,000.

Reduced form / ITT:

$$\rho = 60,000 - 55,000 = 5,000.$$

Winning the lottery raises salary by \$5,000 on average.

First stage:

$$\pi = 0.70 - 0.20 = 0.50.$$

Winning raises course-taking by 50 percentage points.

Wald / IV:

$$\beta_{IV} = \frac{5,000}{0.50} = 10,000.$$

The course raises wages by \$10,000 (for those who responded to the lottery).

Intuition: why scale up by the first stage?

The reduced form (ITT) measures the effect of the *offer*, not the effect of *actually taking* the course.

The offer raised salary by \$5,000. But only half the winners actually took the course. The other half just happened to win the lottery and didn't use the offer.

If only half the winners benefitted, and those who did got the full effect of taking the course, then:

$$\$5,000 = 0.5 \times (\text{effect of course}) + 0.5 \times 0.$$

So the effect of the course must be \$10,000.

Dividing ITT by the first stage rescales the offer effect into a per-person treatment effect.

Reduced form in a regression-friendly form

Going back to the algebra. Dividing both sides of

$$\beta = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, D)}$$

by $\text{Var}(Z)/\text{Var}(Z) = 1$:

$$\beta = \frac{\text{Cov}(Z, Y)/\text{Var}(Z)}{\text{Cov}(Z, D)/\text{Var}(Z)} = \frac{\rho}{\pi}.$$

This makes the connection to regressions explicit. The slopes ρ and π are just OLS coefficients from running:

$$Y_i = \kappa + \rho Z_i + v_i,$$

$$D_i = \mu + \pi Z_i + \eta_i.$$

Two-Stage Least Squares (2SLS): motivation

The Wald estimator handles the binary case cleanly. But what if:

- Z is continuous (e.g., distance to college)?
- We have multiple instruments?
- We have controls W we want to include?

The general-purpose tool is **Two-Stage Least Squares (2SLS)**.

It works in all of these cases and reduces to the Wald estimator when applicable.

2SLS: the procedure

Stage 1. Regress the endogenous variable on the instrument(s) and any controls:

$$D_i = \mu + \pi Z_i + W_i' \psi + \eta_i.$$

Compute the fitted values $\hat{D}_i$.

Stage 2. Regress the outcome on the fitted values and the same controls:

$$Y_i = \alpha + \beta \hat{D}_i + W_i' \phi + u_i.$$

The coefficient $\hat{\beta}$ from stage 2 is the 2SLS estimate.

2SLS: why does it work?

The key insight: $\hat{D}_i$ is, by construction, a linear combination of Z and W .

Both Z and W are exogenous (by assumption). So $\hat{D}_i$ is exogenous too.

Replacing D_i with $\hat{D}_i$ “washes out” the part of D that was contaminated.

We’ve kept the part of D that comes from Z (clean) and thrown away the part that comes from ε (dirty).

Regressing Y on the clean piece gives a consistent estimate of β .

2SLS = Wald in the binary case

Suppose Z is binary and there are no controls.

The first stage gives $\hat{D}_i$:

- For $Z = 1$: $\hat{D} = \mathbb{E}[D|Z = 1]$.
- For $Z = 0$: $\hat{D} = \mathbb{E}[D|Z = 0]$.

$\hat{D}$ only takes two values.

A regression of Y on $\hat{D}$ is just a regression of Y on a two-valued indicator. Its slope is exactly

$$\frac{\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0]}{\mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0]},$$

which is the Wald estimator.

2SLS is a generalisation, not a different idea.

Practical warning: don't run 2SLS by hand

Many software packages let you run the two stages separately. **Don't.**

The standard errors from the second-stage OLS regression are *wrong*.

Why? Because the second stage uses $\hat{D}_i$, which is itself an estimate. OLS doesn't know this and underestimates the variance.

Always use a dedicated command:

- Stata: `ivregress 2sls` or `ivreg2`.
- R: `ivreg()` from AER, or `feols()` from fixest.
- Python: `linearmodels.IV2SLS`.

These packages correctly compute standard errors that account for the first-stage estimation.

Reduced form, first stage, IV: what to report

A well-written IV paper reports all three:

First stage. Shows the instrument is relevant. Report the coefficient $\hat{\pi}$ and the F -statistic.

Reduced form. Shows the instrument affects the outcome. This is what readers can audit even if they question your IV assumptions.

IV / 2SLS. The causal effect. Equal to reduced form / first stage.

Reporting all three is good practice. If the IV looks weird, you can diagnose whether the issue is in the numerator or denominator.

Block 3 wrap-up

- The IV estimator is $\text{Cov}(Z, Y)/\text{Cov}(Z, D)$.
- Equivalently: reduced form divided by first stage.
- 2SLS is the general implementation, including for continuous instruments and controls.
- Always use dedicated software — never run the two stages by hand.

Next: when do the IV assumptions fail, and how do we recognise it?

When IV fails. Independence and exclusion.

Two ways IV can fail

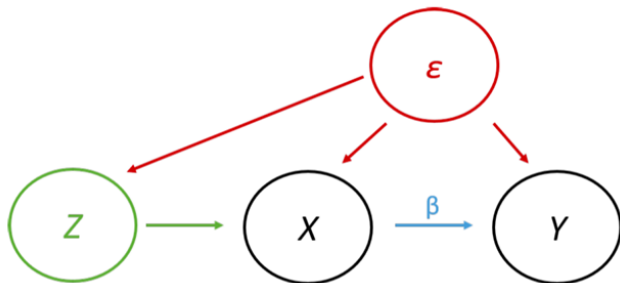
For Z to be a valid instrument:

- It must not be confounded with the unobservables. (**Independence.**)
- It must not affect Y through any path except through D . (**Exclusion.**)

These two are conceptually distinct. Even if your instrument is randomly assigned (independence holds), it can still fail exclusion. And vice versa.

You must argue both, separately, every time.

Threat 1: Independence fails



What this DAG is showing

A red arrow runs from ε to Z .

This means: the unobserved confounders affect the instrument itself.

Now Z is not as-good-as-random. People with high ε are more likely to have high Z .

Result: $\text{Cov}(Z, \varepsilon) \neq 0$, and the IV estimator is biased.

We call this a failure of **independence**.

Independence failure — example: parental education

A researcher wants to estimate the return to a person's own education.

They propose: use *parental* education as an instrument for *own* education.

Check the conditions:

- **Relevance.** Educated parents have more educated kids. Yes.
- **Independence.** Hmm.

Why parental education fails independence

What's in the unobservables for the child?

- Innate ability.
- Family income and wealth.
- Family connections to good employers.

Are any of these correlated with parental education?

Yes. Educated parents are also richer, better-connected, and (genetically and culturally) tend to have more able children.

So the instrument is correlated with the things in the error term. Independence fails.

Independence failure — example: distance to college

Card (1995) used distance to the nearest college as an instrument for years of schooling.

- **Relevance.** Living near a college lowers the cost of attending. Yes.
- **Independence.** Debated.

Why is independence debated? People who live near colleges are different on average:

- Their families chose to settle in areas with strong labor markets.
- These areas have richer industries, higher wages, more amenities.

So distance to college may be correlated with family background — which is in the error term.

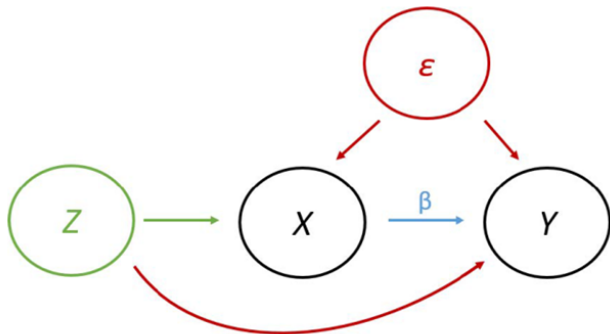
Independence failure — example: police presence

We want to estimate the effect of police on crime. We try using past crime rates as an instrument for current police hiring.

- **Relevance.** High past crime $\rightarrow$ more police hired now. Yes.
- **Independence.** Past crime is determined by past confounders, many of which persist into the present. Almost certainly fails.

This is a typical case where reverse causality / persistence makes the obvious “instrument” actually endogenous.

Threat 2: Exclusion fails



What this DAG is showing

The new red arrow goes *directly* from Z to Y .

This means Z has an effect on Y that does *not* flow through D .

Result: even if Z is randomly assigned (no arrow from ε to Z), it now has two paths to Y :

- $Z \rightarrow D \rightarrow Y$ (the path we want).
- $Z \rightarrow Y$ (the path we don't want).

We can't separate them in the data. The IV estimator picks up both.

Exclusion failure — example: lottery and motivation

Back to the course lottery.

Suppose losing the lottery makes students disappointed and demotivated. They put less effort into other classes and other applications.

Then the lottery affects wages through:

- The course (the channel we want).
- Motivation (the channel we don't want).

The lottery is no longer “excluded” from the wage equation.

$\text{Cov}(Z, \varepsilon) \neq 0$ via the motivation channel. IV is biased.

Exclusion failure — example: quarter of birth

Angrist and Krueger (1991) used quarter of birth as an instrument for schooling.

The mechanism: school-entry rules and compulsory schooling laws mean Q1 vs Q4 birth changes how much school you complete.

But wait. Quarter of birth also affects:

- Age relative to classmates (older or younger).
- Season of labour-market entry.
- Cohort composition.

Any of these could affect wages directly. If so, exclusion fails.

This is a long-debated point about Angrist–Krueger.

Exclusion failure — example: distance to college, again

Even if distance to college is randomly assigned (which it isn't, but for argument), exclusion could still fail.

Why? Local labour markets near colleges may pay higher wages *regardless of one's education*, because:

- More service jobs.
- Better infrastructure.
- Higher overall productivity.

So distance to college affects wages even for people who don't attend.

That's a violation of exclusion. The instrument carries information about wage levels through channels other than the education it instruments for.

Independence vs exclusion: keep them straight

Independence. The instrument is unrelated to confounders. Question to ask: *Why does Z vary across people? Is the assignment random or as-good-as?*

Exclusion. The instrument affects Y only through D . Question to ask: *Holding D fixed, would changing Z change Y ?*

Two different defences for two different threats.

What about relevance? Can it fail?

Yes. If $\text{Cov}(Z, D) = 0$, the IV estimator is undefined (we'd divide by zero).

In practice, the worry is not zero correlation but weak correlation: a **weak instrument**.

A weak instrument causes:

- Imprecise estimates (huge confidence intervals).
- In finite samples, bias toward the (biased) OLS estimate.

We'll discuss diagnostics for this in Block 7.

The level-of-variation question

A practical question: at what level should my instrument vary?

General rule. The instrument generates exogenous variation at the level at which it varies. That level must be at least as coarse as the endogenous variable.

Example. Door-to-door campaign assigned at the neighbourhood level. The instrument Z_{nt} varies at the neighbourhood-time level. The treatment D_{ihnt} (individual service take-up) varies at the individual level.

This is fine. The campaign was assigned to neighbourhoods, not individuals. The variation we exploit is between neighbourhoods.

Two practical consequences:

- Cluster standard errors at the instrument's level (neighbourhood-time, here).
- The exogeneity argument applies at that level too.

When the level is suspicious

If your instrument varies at a *finer* level than the treatment, ask why.

Example: D is whether a household uses a service. Z is whether individual i in the household was contacted by an outreach worker.

Why was individual i contacted and not their spouse? If the worker self-selected to contact the more interested family member, Z is no longer random within the household. Independence fails.

Safe configurations:

- Z and D at the same level (course lottery).
- Z coarser than D , with Z assigned by institutional rule at the coarser level (neighbourhood, school, district, year, cohort).

Block 4 wrap-up

- Independence and exclusion are separate threats.
- Independence: is Z as-good-as-random?
- Exclusion: does Z affect Y only through D ?
- Both must be defended on institutional grounds. Data alone can't decide.
- The level at which Z varies determines the level of the identification argument and clustering.

What if everyone has a different treatment effect?

So far we've assumed a single β

The model

$$Y_i = \alpha + \beta D_i + \varepsilon_i$$

assumes one number β describes the causal effect for everyone.

Is this realistic?

Why a single β is not realistic

- The effect of a year of schooling differs across people: gains are larger for some than others.
- The effect of a medication differs across patients: some respond, some don't.
- The effect of an anti-poverty programme differs across families.

Real treatment effects are **heterogeneous**.

Each person has their own β_i .

The model becomes:

$$Y_i = \alpha + \beta_i D_i + \varepsilon_i.$$

Potential outcomes: the framework

A cleaner way to write heterogeneous effects uses **potential outcomes**.

For each individual i , imagine two parallel worlds:

- $Y_i(1)$: the outcome if i is treated ($D_i = 1$).
- $Y_i(0)$: the outcome if i is untreated ($D_i = 0$).

The causal effect on i is:

$$\beta_i = Y_i(1) - Y_i(0).$$

In any single dataset, we observe only one of $Y_i(0)$ or $Y_i(1)$ for each person. The other is counterfactual — it didn't happen, but we can imagine it.

Different population averages

With heterogeneous effects, “the” treatment effect is no longer a single number. There are several different averages we might care about:

- **ATE**: the average treatment effect in the whole population, $\mathbb{E}[\beta_i]$.
- **ATT**: the average effect on the treated, $\mathbb{E}[\beta_i \mid D_i = 1]$.
- **LATE**: the average effect for a specific subgroup — coming up.

With homogeneous effects all of these are the same number. With heterogeneous effects they differ.

The new question

With heterogeneous effects, what does our IV estimator recover?

It's not the same β as before, because there is no single β .

The answer (Imbens & Angrist 1994): under stronger assumptions, IV recovers the **Local Average Treatment Effect (LATE)**.

LATE = the average treatment effect for a particular subpopulation — the people whose behavior was changed by the instrument.

We'll develop this slowly.

Potential outcomes for D too

We need not just potential outcomes for Y , but also potential *treatments*.

For each individual i :

- $D_i(1)$: treatment i would receive if assigned $Z_i = 1$.
- $D_i(0)$: treatment i would receive if assigned $Z_i = 0$.

With a binary instrument:

$$D_i = Z_i \cdot D_i(1) + (1 - Z_i) \cdot D_i(0).$$

We see one of $D_i(0)$ or $D_i(1)$, not both. The unobserved one is counterfactual.

Example: course lottery

For a given student i :

- $D_i(1)$: would she enroll if she *won* the lottery?
- $D_i(0)$: would she enroll if she *lost* the lottery?

For some students:

- $D_i(1) = 1, D_i(0) = 0$: only enrolls if she wins.
- $D_i(1) = 1, D_i(0) = 1$: enrolls regardless (maybe took the course already).
- $D_i(1) = 0, D_i(0) = 0$: doesn't enroll either way.

The instrument moves *some* people but not all.

The four IV assumptions, restated

For IV under heterogeneous effects we need four conditions:

- 1 **Independence (random assignment).** Z is unrelated to potential outcomes and potential treatments.
- 2 **Exclusion.** The instrument affects Y only through D .
- 3 **First stage.** $\mathbb{E}[D_i(1) - D_i(0)] \neq 0$. The instrument moves treatment on average.
- 4 **Monotonicity.** The instrument moves everyone in the same direction (no defiers).

Independence and exclusion are the heterogeneous-effects version of what we had before. First-stage is just relevance restated. Monotonicity is new — we'll motivate it carefully.

Independence, formally

Statement. $\{Y_i(d, z), D_i(z)\} \perp\!\!\!\perp Z_i$.

In words: the instrument is statistically independent of everyone's potential outcomes and potential treatments.

This is the heterogeneous version of “ Z is as-good-as-randomly assigned.”

What it buys us: we can compute means of Y and D by Z -status, and these means represent the true potential-outcome distributions in each Z -group.

Threats. Anything that determines Z also affects Y or D directly.

Independence in the course-lottery example

The lottery is run by computer. There is no relationship between lottery outcomes and anything about the students.

Z is literally a coin flip (well, a number from a random draw).

So winning vs losing is independent of the student's ambition, ability, family background, etc.

Independence holds by design.

Random assignment is the gold standard for satisfying independence.

Exclusion, formally

Statement. $Y_i(d, 1) = Y_i(d, 0)$ for all d .

In words: holding the actual treatment d fixed, the value of the instrument has no further effect on Y .

This lets us drop the second argument and just write $Y_i(d)$.

The exclusion restriction is a structural claim about the world. It says: “The only way Z matters for Y is by changing D .”

Exclusion in the lottery example

The lottery is excluded from the wage equation if:

- Winning a course slot doesn't directly affect your wages.
- Losing a course slot doesn't directly affect your wages.
- The only thing that matters is whether you actually took the course.

Plausible? Mostly yes. But subtle threats exist:

- Lottery losers might be demoralised (mentioned earlier).
- Lottery winners might tell employers "I'm in the X programme" even if they don't take the course, getting them a signaling benefit.

These are exclusion violations, not independence violations.

Independence vs exclusion: same example

For the same lottery, the two threats are different:

Independence threat. The lottery isn't really random. Maybe the registrar rigged it. (Unlikely here.)

Exclusion threat. The lottery itself, by being a lottery, changes behaviour even of people who don't take the course. (Plausible.)

Both attack IV validity, but in different ways. You need to argue them separately.

First stage, formally

Statement. $\mathbb{E}[D_i(1) - D_i(0)] \neq 0$.

In words: on average, switching Z from 0 to 1 changes treatment status.

This is the heterogeneous-effects analogue of “relevance.”

Crucially, this assumption is testable. Just run the first-stage regression and check whether the coefficient on Z is significantly different from zero.

The first-stage F -statistic is the standard diagnostic. We'll see thresholds in Block 7.

Monotonicity, formally

Statement. $D_i(1) \geq D_i(0)$ for all i (or $D_i(1) \leq D_i(0)$ for all i).

In words: the instrument moves everyone in the same direction.

For a lottery: winning never makes you *less* likely to take the course than losing.

This rules out a particular kind of behaviour: contrarians who do the opposite of what the instrument suggests.

Why monotonicity matters

Without monotonicity, the instrument could push some people into treatment and other people out.

These two groups' effects would cancel out (with opposite signs) in the IV estimator.

The result could be zero even if the treatment effect is large for everyone.

Monotonicity rules out this kind of cancellation by saying nobody is moved in the “wrong” direction.

Monotonicity in plain English

Monotonicity says: “If you’d take the treatment without being encouraged, you’d also take it when you are encouraged.”

Or equivalently: “Encouragement never backfires.”

For a lottery offer to take a course: monotonicity says there’s nobody so contrarian that getting the offer makes them *less* likely to take the course.

For most natural experiments this is plausible. But it’s a substantive assumption, not a tautology.

Block 5 wrap-up

- Treatment effects are heterogeneous.
- We use potential outcomes for both Y and D .
- Four assumptions: independence, exclusion, first stage, monotonicity.
- Independence and exclusion replace the OLS-IV exogeneity condition.
- First stage replaces relevance.
- Monotonicity is new — it rules out defiers.

Next: under these four, what exactly does IV identify?

The four response types and the LATE theorem.

Sorting people by their $(D_i(0), D_i(1))$ pair

Each person has a pair $(D_i(0), D_i(1))$.

With binary treatment, each $D_i(z)$ is 0 or 1. So the pair takes one of four possible values.

These four possibilities define the **response types**:

	$D_i(0) = 0$	$D_i(0) = 1$
$D_i(1) = 1$	Complier	Always-taker
$D_i(1) = 0$	Never-taker	Defier

Type 1: Always-takers

Always-takers: $D_i(0) = D_i(1) = 1$.

These are people who take the treatment regardless of the instrument.

Course-lottery example: students who would enroll in the course whether or not they won the lottery. Maybe they have a connection to the professor, are on a waitlist, or it's required for their program.

Important: “always-taker” is jargon. It does NOT mean “always takes any treatment.” It just means “takes *this* treatment regardless of *this* instrument.”

Type 2: Never-takers

Never-takers: $D_i(0) = D_i(1) = 0$.

These are people who never take the treatment, regardless of the instrument.

Course-lottery example: students who would not enroll either way. Maybe they have a schedule conflict, the course is irrelevant to their plans, or they just don't want to take it.

Same jargon warning: “never-taker” refers strictly to behavior under *this* instrument.

Type 3: Compliers

Compliers: $D_i(0) = 0$, $D_i(1) = 1$.

These are people who take the treatment if and only if the instrument tells them to.

Course-lottery example: students who would enroll if they won the lottery and not enroll if they lost. They are responsive to the offer. They don't have the connections that always-takers do, but they are interested enough to take advantage of a slot.

This is the group whose treatment effect we will identify.

Type 4: Defiers

Defiers: $D_i(0) = 1, D_i(1) = 0$.

These are people who take the treatment if the instrument tells them *not* to.

Course-lottery example: students who would enroll only if they *lost* the lottery (somehow). This is hard to imagine.

Monotonicity is precisely the assumption that there are no defiers.

Why monotonicity rules out defiers

Recall: monotonicity says $D_i(1) \geq D_i(0)$ for everyone.

- Always-takers: $D(1) = 1 \geq 1 = D(0)$. ✓
- Never-takers: $D(1) = 0 \geq 0 = D(0)$. ✓
- Compliers: $D(1) = 1 \geq 0 = D(0)$. ✓
- Defiers: $D(1) = 0 < 1 = D(0)$. **Violates.**

So under monotonicity, the population consists of three types: always-takers, never-takers, and compliers.

Worked example 1: school voucher lottery

Z = won a private-school voucher; D = attends private school.

- **Always-takers.** Wealthy families who would send their kids to private school regardless. They'd pay tuition out of pocket.
- **Never-takers.** Families committed to the public school. Maybe the nearest private school is too far; maybe they prefer public on principle.
- **Compliers.** Families on the margin who use the voucher to enroll.
- **Defiers.** Families who'd attend private school only if they lost. Implausible.

The IV estimate from this lottery is the effect of private schooling *for compliers*.

Worked example 2: Vietnam draft lottery

Angrist (1990). Z = low draft number (eligible for draft); D = served in the military.

- **Always-takers.** Voluntary enlistees. Would serve regardless.
- **Never-takers.** Avoid service even if drafted. Medical exemptions, college deferments, conscientious objectors.
- **Compliers.** Served only because they were drafted.
- **Defiers.** Would serve only if not drafted. Implausible.

The IV estimate is the effect of military service *for compliers*: men who served only because of the draft.

Worked example 3: quarter of birth

Angrist–Krueger. $Z = \text{Q4 birth (vs Q1)}$; $D = \text{completed high school}$.

Mechanism: Q4-born students enter school younger and so must stay longer to satisfy compulsory schooling age.

- **Always-takers.** Students who'd finish high school regardless (planning to attend college).
- **Never-takers.** Students who'd drop out as soon as legal age allows, regardless of when they entered.
- **Compliers.** Students at the compulsory-schooling boundary. Q4 birth keeps them in a little longer.
- **Defiers.** Students who'd stay in school only if they could legally drop out earlier. Implausible.

Worked example 4: OHIE Medicaid lottery

Finkelstein et al. (2012). Z = won Medicaid lottery; D = enrolled in Medicaid.

- **Always-takers.** People who would find Medicaid via another route (disability application, etc.) even without the lottery.
- **Never-takers.** Won the lottery but didn't complete the application. Maybe paperwork too hard, language barrier, or simply changed their mind.
- **Compliers.** Enrolled because they won.
- **Defiers.** Would enroll only if they lost. Implausible.

In OHIE the first stage is about 0.25 — only 25% of lottery winners actually enrolled. So compliers are a quarter of the lottery population.

Worked example 5: tuition subsidies

Z = randomly offered tuition subsidy; D = attended college.

- **Always-takers.** Would attend college regardless (have rich parents, scholarships, etc.).
- **Never-takers.** Would not attend regardless (have other plans, dropped out earlier, etc.).
- **Compliers.** On the financial margin. The subsidy pushes them in.
- **Defiers.** Implausible.

This is the type of LATE that policy-makers most often want, because it's the effect of a marginal tuition change.

The LATE theorem

Imbens & Angrist (1994)

Under the four assumptions (independence, exclusion, first stage, monotonicity), the IV / Wald estimator identifies:

$$\beta_{IV} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{Complier}].$$

This is the **Local Average Treatment Effect (LATE)**.

In words: IV recovers the average treatment effect for compliers — the subgroup whose behaviour was changed by the instrument.

The next slides walk through the proof.

LATE proof: step 1, the ITT decomposition

By independence:

$$\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0] = \mathbb{E}[Y(D(1), 1) - Y(D(0), 0)].$$

Decompose by response type using the law of total expectation:

$$\begin{aligned}\mathbb{E}[Y(D(1), 1) - Y(D(0), 0)] &= \mathbb{E}[Y(D(1), 1) - Y(D(0), 0)|\text{Comp.}] \cdot P(\text{Comp.}) \\ &\quad + \mathbb{E}[Y(D(1), 1) - Y(D(0), 0)|\text{NT}] \cdot P(\text{NT}) \\ &\quad + \mathbb{E}[Y(D(1), 1) - Y(D(0), 0)|\text{AT}] \cdot P(\text{AT}) \\ &\quad + \mathbb{E}[Y(D(1), 1) - Y(D(0), 0)|\text{Def.}] \cdot P(\text{Def.}).\end{aligned}$$

(NT = never-taker, AT = always-taker.)

Now we simplify each term.

LATE proof: step 2, apply exclusion

By exclusion, $Y(d, 1) = Y(d, 0) = Y(d)$. So the difference inside each expectation simplifies:

- **Compliers.** $D(0) = 0, D(1) = 1$, so the difference is $Y(1) - Y(0) =$ the treatment effect.
- **Never-takers.** $D(0) = D(1) = 0$, so the difference is $Y(0) - Y(0) = 0$.
- **Always-takers.** $D(0) = D(1) = 1$, so the difference is $Y(1) - Y(1) = 0$.
- **Defiers.** $D(0) = 1, D(1) = 0$, so the difference is $Y(0) - Y(1) = -(Y(1) - Y(0))$.

Two terms vanish (never-takers and always-takers contribute nothing).

LATE proof: step 3, apply monotonicity

By monotonicity, $P(\text{Defier}) = 0$.

So the defier term drops out entirely.

What remains:

$$\mathbb{E}[Y|Z = 1] - \mathbb{E}[Y|Z = 0] = \mathbb{E}[Y(1) - Y(0)|\text{Complier}] \cdot P(\text{Complier}).$$

That is: the ITT (reduced form) equals the treatment effect for compliers, scaled by the share of compliers.

LATE proof: step 4, the denominator

By independence, $\mathbb{E}[D|Z = z] = \mathbb{E}[D(z)]$.

So:

$$\mathbb{E}[D|Z = 1] - \mathbb{E}[D|Z = 0] = \mathbb{E}[D(1) - D(0)].$$

Under monotonicity, $D(1) - D(0)$ is 0 for non-compliers and 1 for compliers. So:

$$\mathbb{E}[D(1) - D(0)] = P(\text{Complier}).$$

The first stage equals the share of compliers.

LATE proof: step 5, divide

Putting it all together:

$$\beta_{IV} = \frac{\mathbb{E}[Y|Z=1] - \mathbb{E}[Y|Z=0]}{\mathbb{E}[D|Z=1] - \mathbb{E}[D|Z=0]} = \frac{\mathbb{E}[Y(1) - Y(0)|\text{Comp.}] \cdot P(\text{Comp.})}{P(\text{Comp.})}.$$

The $P(\text{Complier})$ cancels:

$$\beta_{IV} = \mathbb{E}[Y(1) - Y(0)|\text{Complier}].$$

Done. IV identifies the average treatment effect for compliers. This is the LATE.



Why “local” ?

LATE = Local Average Treatment Effect.

“Local” refers to a *subgroup of people*, not a geographical area.

The subgroup is: compliers — those whose behaviour responds to this particular instrument.

A different instrument → a different set of compliers → a different LATE.

This is a feature, not a bug. LATE is honest about which population the estimate applies to.

Identifying type shares (but not who is who)

We can never tell who is which type from the data. But we can compute the shares.

Under monotonicity, no defiers:

$$P(\text{Always-taker}) = P(D = 1|Z = 0).$$

$$P(\text{Never-taker}) = P(D = 0|Z = 1).$$

$$P(\text{Complier}) = P(D = 1|Z = 1) - P(D = 1|Z = 0).$$

The complier share IS the first stage.

A bigger first stage $\rightarrow$ more compliers $\rightarrow$ LATE applies to a larger group.

Why can't we identify individuals?

Look at a person with $Z = 1$ and $D = 1$. Are they a complier or an always-taker?

Both possible:

- Compliers with $Z = 1$ have $D = D(1) = 1$.
- Always-takers always have $D = 1$.

We can't tell from observed data which they are.

Similarly, someone with $Z = 0$ and $D = 0$ could be a complier or a never-taker.

We can only identify averages within types, not individual type membership.

Numerical example: type shares

Suppose:

- $P(D = 1|Z = 1) = 0.7$.
- $P(D = 1|Z = 0) = 0.2$.

Then:

- Always-takers: $P(D = 1|Z = 0) = 0.2$. So 20% of the population.
- Never-takers: $P(D = 0|Z = 1) = 1 - 0.7 = 0.3$. So 30%.
- Compliers: $0.7 - 0.2 = 0.5$. So 50%.

The LATE applies to that 50% of the population — the compliers.

LATE is policy-relevant when the instrument is

Different instruments produce different LATEs because they shift different compliers.

This means: when interpreting an IV estimate, you should ask which compliers the instrument selected.

Example. Two different IVs for the return to education:

- Vietnam draft lottery: compliers are men who got more schooling because they got deferments instead of being drafted.
- Distance to college: compliers are people who attended college because it was nearby.

Different groups, different LATEs. Both valid, both meaningful, but applying to different subpopulations.

Oreopoulos (2006): when LATE approaches ATE

Oreopoulos used the 1947 UK reform that raised the minimum school-leaving age from 14 to 15.

This affected *nearly half* of all 14-year-olds nationwide.

Why does this matter? Because the complier group is a huge fraction of the population.

When compliers are most of the population, the LATE is close to the population ATE.

Lesson: the more universal the instrument, the more the LATE looks like an ATE.

Block 6 wrap-up

- Each person has a response type: complier, always-taker, never-taker, or defier.
- Monotonicity rules out defiers.
- Under all four assumptions, IV identifies the LATE: the average treatment effect for compliers.
- We can compute type *shares* but never type *membership* for individuals.
- Different instruments produce different LATEs.

Next: how do we know if our instrument is strong enough to trust?

Weak instruments, overidentification, and standard errors.

The weak-instrument problem

The first stage condition says $\text{Cov}(Z, D) \neq 0$.

But this is a knife-edge condition. What if $\text{Cov}(Z, D)$ is non-zero but *small*?

This is the **weak instrument** problem.

Even though IV is technically consistent, in practice things go badly.

What goes wrong with weak instruments (1)

Wide confidence intervals.

The IV estimator is ρ/π , where π is the first stage.

If π is small, the denominator is small. Small sampling variation in the numerator gets amplified.

Confidence intervals become enormous.

What goes wrong with weak instruments (2)

Finite-sample bias toward OLS.

In finite samples, $\hat{\pi}$ has some sampling error. This error is correlated with the residuals from the first stage, which are themselves correlated with the structural errors.

Result: IV gets pulled toward the (biased) OLS estimate.

With very weak instruments, IV can be *more biased* than OLS.

Counterintuitive, but true.

The bias-amplification formula

If exclusion is slightly violated ($\text{Cov}(Z, \varepsilon)$ slightly non-zero):

$$\text{plim } \hat{\beta}_{\text{IV}} = \beta + \frac{\text{Cov}(Z, \varepsilon)}{\text{Cov}(Z, D)}.$$

Notice the denominator. If $\text{Cov}(Z, D)$ is small, the bias term explodes.

Lesson. A weak, slightly endogenous instrument can produce a wildly biased estimate.

The combination of “weak” and “slightly bad” is much worse than either alone.

The first-stage F -statistic

Standard diagnostic: the F -statistic from the first-stage regression, testing whether the instrument coefficient is zero.

Run:

$$D_i = \mu + \pi Z_i + W_i' \psi + \eta_i.$$

The F -statistic is testing $H_0 : \pi = 0$.

A small F means we can't reject "the instrument doesn't matter."

A large F means we can reject it confidently.

Thresholds for the F -statistic

Classic rule of thumb (Staiger-Stock 1997). $F > 10$.

This was the gold standard for decades. Below 10, worry. Above 10, you're fine.

Stock-Yogo (2005). Refined this with formal tables based on tolerable bias relative to OLS, and number of instruments.

Lee-McCrary-Moreira-Porter (2022). Recent work shows the $F > 10$ rule does *not* give correct size for standard t -tests in finite samples.

To use $|t| > 1.96$ as a valid 5% test, you need $F > 104.7$.

This is a striking result. The conservative approach: report F , and if it's not huge, use weak-instrument-robust inference.

Numerical illustration of weak-instrument bias

Suppose:

- True $\beta = 1$.
- $\text{Cov}(Z, \varepsilon) = 0.01$ (tiny exclusion violation).

Case A. Strong instrument: $\text{Cov}(Z, D) = 0.5$.

$$\text{Bias} = 0.01/0.5 = 0.02.$$

Estimate is 1.02. Fine.

Case B. Weak instrument: $\text{Cov}(Z, D) = 0.02$.

$$\text{Bias} = 0.01/0.02 = 0.5.$$

Estimate is 1.5. A 50% error.

Same tiny exclusion violation, vastly different consequences.

Overidentification: when you have multiple instruments

Sometimes you have more instruments than endogenous regressors. Example: two different lotteries, both for the same course.

This is called **overidentification**.

The system is “over-identified” because each instrument independently identifies β . If both instruments are valid, both should give the same answer.

If they don't, something is wrong.

The Sargan-Hansen J -test

The Sargan-Hansen J -test asks: are all instruments mutually consistent?

Under the null that all instruments are exogenous, the 2SLS residuals should be uncorrelated with every instrument.

The test statistic is nR^2 from regressing 2SLS residuals on all instruments. Distributed χ^2_{r-k} where r = number of instruments and k = number of endogenous variables.

Caveats of the overidentification test

What rejection means. At least one instrument is bad. Useful.

What failure to reject means. The instruments are mutually consistent. *Not* that they are all valid.

They could all be biased in the same direction. The test doesn't catch that.

Exactly identified case. If $r = k$, you can't run the test. The argument for exclusion must be made on institutional grounds, not statistical ones.

This is why economists generally prefer one strong, defensible instrument over many shaky ones.

Standard errors: cluster at the level of the instrument

Conventional OLS standard errors assume each observation is an independent draw.

If your instrument varies at the neighbourhood level, then all observations within the same neighbourhood share the same Z value. They are not independent.

Rule. Cluster standard errors at the level at which the instrument varies.

If Z varies at the school level: cluster at the school.

If Z varies at the cohort-year level: cluster at the cohort-year.

This is one of the most common mistakes in applied IV work.

Block 7 wrap-up

- Weak instruments cause wide confidence intervals and finite-sample bias.
- Report the first-stage F -statistic. Old threshold: 10. Modern threshold (for valid t -tests): much higher.
- Bias amplification: weak + slightly endogenous = catastrophic.
- Overidentification tests can flag bad instruments but can't certify good ones.
- Cluster standard errors at the level of the instrument.

Five worked examples. Each tells a story.

Application 1: Oregon Health Insurance Experiment

Question. Does Medicaid coverage improve health and reduce financial strain?

Setting. In 2008 Oregon opened a limited number of Medicaid slots for low-income adults. Too many applied. Slots were allocated by lottery.

Variables.

- Z_i : won the lottery.
- D_i : actually enrolled in Medicaid.
- Y_i : hospital admissions, out-of-pocket spending, self-reported health.

OHIE: the structure

The first stage is about 0.25. Only one in four lottery winners actually enrolled. (Application paperwork was burdensome.)

The reduced-form effect on hospital admissions: 0.5 percentage points.

The IV estimate: $0.5/0.25 = 2$ percentage points. Medicaid raises hospital admission probability by 2 pp *for compliers*.

OHIE: who are the compliers?

- **Always-takers.** People who would find Medicaid via another channel (disability claim, special program). Not in our complier group.
- **Never-takers.** Won the lottery but didn't complete the application. Many people in this group. Not in our complier group.
- **Compliers.** Won the lottery and enrolled because of it. About 25% of lottery participants.

The LATE applies to this 25%.

For policy purposes, this is exactly the right population if you're considering expanding Medicaid through an automated process that requires people to apply.

Application 2: Angrist (1990), Vietnam draft

Question. What's the causal effect of military service on civilian earnings?

Setting. Vietnam-era draft lottery. Men with low lottery numbers were more likely to be drafted.

Variables.

- Z_i : lottery number was below the cutoff.
- D_i : served in the military.
- Y_i : civilian earnings 10+ years later.

The lottery is essentially random. Independence is trivially satisfied.

Angrist (1990): findings

The IV estimate: white veterans of Vietnam earned roughly 15% less than otherwise-similar non-veterans.

Naïve OLS comparing veterans to non-veterans would have given different answers, because veteran status is correlated with background.

Who are the compliers? Men who served because they were drafted. The LATE is the effect of military service on *these* men's earnings.

Not the effect on voluntary enlistees, who chose service for their own reasons.

Application 3: Angrist-Krueger (1991), quarter of birth

Question. What's the return to a year of schooling?

Setting. Compulsory schooling laws say you must stay in school until age 16. Q1-born students enter school old and reach 16 sooner. Q4-born students enter school young and reach 16 having completed more years.

Variables.

- Z_i : quarter of birth (e.g., Q4 vs Q1).
- D_i : years of schooling.
- Y_i : log earnings.

Angrist-Krueger: findings and controversy

The IV estimates of the return to schooling were about 10% per year, similar to OLS.

Implication: ability bias might not be as severe as feared.

But. The instrument has been contested:

- Exclusion: quarter of birth affects more than just schooling. Age-at-test, cohort composition.
- Weak instrument: first stage is small. Bound-Jaeger-Baker (1995) showed serious finite-sample bias.

Even the most famous applications have honest critics.

Application 4: Card (1995), distance to college

Question. Same as above: return to a year of schooling.

Setting. Some young people grew up near a 4-year college; others did not.

Variables.

- Z_i : lived near a 4-year college at age 14.
- D_i : years of schooling.
- Y_i : log earnings.

Card (1995): findings and controversy

IV estimates: roughly 14% return per year of schooling. *Higher* than OLS.

This is puzzling: standard ability bias should make OLS higher.

Card's interpretation: the LATE is for a particular complier group (kids whose schooling decision was on the financial margin), and they have especially high returns.

Alternative: attenuation bias in measured schooling drove OLS down; IV corrects for it.

Controversy. Distance to college may not be independent: families self-select into college towns.

Application 5: Kline & Walters (2016), Head Start

Question. Does Head Start improve test scores for disadvantaged children?

Setting. The Head Start Impact Study (HSIS) randomised offers to families.

New wrinkle. Treatment is no longer binary.

- $D = h$: Head Start.
- $D = c$: other preschool centre.
- $D = n$: no preschool.

About 1/3 of control families enrolled in *other* preschools.

Head Start: the substitution problem

The simple Wald estimator using $\mathbf{1}\{D = h\}$ identifies a *mixture* of two effects:

- Effect of Head Start vs other preschool (*c*-compliers).
- Effect of Head Start vs no preschool (*n*-compliers).

If compliers are mostly drawn from other preschools, the IV gives a small number even if Head Start works.

This is called **substitution bias** (Heckman et al. 2000).

Kline-Walters: their method, in spirit

They use additional structure to back out separate sub-LATEs:

- $LATE_{nh}$: effect of Head Start vs. staying home.
- $LATE_{ch}$: effect of Head Start vs. other preschool.

The overall Wald is:

$$LATE_h = S_c \cdot LATE_{ch} + (1 - S_c) \cdot LATE_{nh}.$$

Their estimates:

- $LATE_{nh} \approx 0.37$ (large gains for kids drawn from home).
- $LATE_{ch} \approx 0$ (no effect over other preschools).

Conclusion. Head Start has big effects for kids who'd otherwise stay home — but the average looks small because many compliers were already in other preschools.

Fuzzy RD as IV

A related design: fuzzy regression discontinuity.

Sharp RD. Treatment is determined by crossing a cutoff. Everyone above the cutoff is treated; nobody below.

Fuzzy RD. Crossing the cutoff changes the *probability* of treatment but not deterministically.

The cutoff indicator is an instrument for treatment status. Fuzzy RD = IV.

Fuzzy RD: example from past exams

SEMUJERES CDMX. Door-to-door campaign assigned to certain neighbourhoods.

- Running variable: distance to the boundary of a targeted neighbourhood.
- Cutoff: distance = 0 (the boundary itself).
- Treatment: service take-up.
- Outcome: probability of a restraining order issued.

Crossing the boundary jumps the probability of campaign exposure $\rightarrow$ jumps the probability of service take-up. Fuzzy because not 100% compliance.

The indicator $\mathbf{1}\{r \geq 0\}$ is the instrument.

Fuzzy RD: empirical model

$$\text{First stage: } D_i = \pi_0 + \pi_1 \mathbf{1}\{r_i \geq 0\} + f_D(r_i) + \eta_i,$$

$$\text{Reduced form: } Y_i = \rho_0 + \rho_1 \mathbf{1}\{r_i \geq 0\} + f_Y(r_i) + v_i,$$

$$\text{2SLS: } Y_i = \alpha + \beta \hat{D}_i + h(r_i) + u_i.$$

f_D, f_Y, h are flexible (usually local-linear) functions of the running variable.

Estimate on a narrow window around the cutoff. Calonico-Cattaneo-Titiunik methods pick the bandwidth automatically.

Cluster standard errors at the level of the running variable (neighbourhood here).

Reading a fuzzy-RD/IV output table

Past-exam table reports:

- Coefficient on the endogenous variable: $\hat{\beta}$.
- Standard error.
- First-stage F p-value: indicates instrument strength.
- Local polynomial order: linear, etc.
- Bandwidth: how close to the cutoff.
- Mean of dependent variable in control: to compare magnitudes.

What to say when interpreting:

- 1 Is the first stage strong? Report the F or its p-value.
- 2 Is the coefficient statistically significant?
- 3 Substantive size: compare to the control-group mean.
- 4 Whose LATE is this?
- 5 What threats remain (exclusion, independence)?

Worked exam interpretation

From a past exam: $\hat{\beta} = 0.176$, $SE = 0.031$, first-stage F p-value < 0.0001 , control mean $= 0.407$.

Interpretation.

- Strong first stage (p-value very low).
- $t \approx 5.7$. Highly significant.
- Compliers near the border who take up specialised services are 17.6 percentage points more likely to receive a restraining order.
- Relative to control mean of 40.7%, this is a 43% proportional increase. Large.
- Whose LATE? Women whose service take-up responded to the neighbourhood-level campaign near the border.
- Threats: campaign visibility could deter offenders directly (exclusion); border families could differ from non-border families (independence).

Common pitfalls in writing about IV

Pitfall 1. Saying “my instrument is valid because the F is high.” The F tests relevance, not exogeneity. Exogeneity is not statistically testable.

Pitfall 2. Reporting OLS standard errors after manually running two-stage regressions. Wrong by default.

Pitfall 3. Calling the IV estimate “the ATE.” Under heterogeneity, IV gives a LATE.

Pitfall 4. Using parental education, distance, or other proxies for ability as IVs without considering exclusion.

Pitfall 5. Failing to cluster standard errors at the level of the instrument.

Pitfall 6. Pooling many weak instruments to “get” a higher F . This can amplify bias.

Things we didn't cover (frontier)

- Marginal treatment effects (Heckman & Vytlacil): generalises LATE to continuous-instrument settings.
- Multiple instruments with non-monotone first stages: Mogstad-Torgovitsky-Walters (2021).
- Weak-instrument-robust inference: Anderson-Rubin confidence sets.
- Nonparametric IV.
- IV in nonlinear models (probit, logit, Poisson).

Each is a deeper topic. The basics in this lecture are what you need for past exam questions, replication exercises, and reading most applied papers.

Block 8 wrap-up

- IV recovers causal effects from observational data when a good instrument exists.
- Five canonical applications: OHIE, draft lottery, quarter of birth, distance to college, Head Start.
- Fuzzy RD is IV with the cutoff indicator as instrument.
- Interpret estimates carefully: whose LATE, what threats, what magnitude.

What you should leave with:

- OLS fails under endogeneity. Three sources: OVB, measurement error, simultaneity.
- IV uses a clever third variable to isolate clean variation.
- Two properties: relevance and exogeneity.
- Under homogeneous effects, IV gives a single β .
- Under heterogeneous effects and the four assumptions, IV gives the LATE for compliers.
- Diagnostics: F -statistic, overidentification, clustered standard errors.
- Always argue independence and exclusion separately.